

Zhibing Zhao

Ph.D. in Computer Science

*Research Scientist at Meta
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Research Interests

- Machine Learning
- Time Series
- Ads, Search, and Recommendation
- Natural Language Processing

Experience

- 2025–Present **Research Scientist**, META, Bellevue, WA, US.
- Click refinement: conversion prediction for high/low intent clicks, achieving statistically significant 0.18% global ads score improvement;
 - Mastercook chef: end-to-end model building, training and launch, achieving statistically significant 0.18% global ads score improvement.
- 2022–2025 **Research Scientist**, BYTEDANCE, Bellevue, WA, US.
- SQL query optimization: novel learning-to-rank approach to achieve 6x speedup;
 - Time series: ensemble learning for storage usage forecasting, saving 10% storage.
- 2020–2022 **Data & Applied Scientist**, MICROSOFT, Bellevue, WA, US.
- Hybrid ranking models for search and page recommendation:
- Built hybrid ranking model with 3.0% gain in NDCG@3 for page recommendation;
 - Built compressed MEB model that has 3.2% gain in NDCG@1 for search.
- 2019 Summer **Data and Applied Scientist Intern**, MICROSOFT, Bellevue, WA, US.
- 2018 Summer **Research Intern**, MICROSOFT RESEARCH ASIA, Beijing, China.
- Award of Excellence*

Education

- 2015–2020 **Ph.D., Computer Science**, Rensselaer Polytechnic Institute, Troy, NY, US.
- 2012–2014 **M.S., Electrical Engineering**, University of Connecticut, Storrs, CT, US.
- 2008–2012 **B.Eng., Electrical Engineering**, Tsinghua University, Beijing, China.

Selected Publications

- Xianghong Xu, **Zhibing Zhao**, Tieying Zhang, Rong Kang, Luming Sun, and Jianjun Chen, "COOOL: A Learning-To-Rank Approach for SQL Hint Recommendations". In 5th International Workshop on Applied AI for Database Systems and Applications (**AIDB 2023**).
- Luming Sun, Shijin Gong, Tieying Zhang, Fuxin Jiang, **Zhibing Zhao**, Jianjun Chen, and Xinyu Zhang, "SUF5: A Generic Storage Usage Forecasting Service Through Adaptive Ensemble Learning". In Proceedings of the 39th IEEE International Conference on Data Engineering (**ICDE 2023**).

- **Zhibing Zhao**, Ao Liu, and Lirong Xia, "Learning Mixtures of Random Utility Models with Features from Top- l Orders". In Proceedings of International Joint Conference on Artificial Intelligence (**IJCAI-22**).
- **Zhibing Zhao**, Yingce Xia, Tao Qin, Lirong Xia, and Tie-Yan Liu, "Dual Learning: Theoretical Study and an Algorithmic Extension", **SN Computer Science**, 2021.
- **Zhibing Zhao** and Lirong Xia, "Learning Mixtures of Plackett-Luce Models from Structured Partial Orders", in *Proceedings of 33rd Conference on Neural Information Processing Systems (NeurIPS-19)*.
- Ao Liu, **Zhibing Zhao**, Chao Liao, Pinyan Lu and Lirong Xia, "Learning Plackett-Luce Mixtures from Partial Preferences", in *Proceedings of 33rd AAAI Conference on Artificial Intelligence (AAAI-19)*.
- Jun Wang, Sujoy Sikdar, Tyler Shepherd, **Zhibing Zhao**, Chunheng Jiang and Lirong Xia, "Practical Algorithms for STV and Ranked Pairs with Parallel Universes Tiebreaking", in *Proceedings of 33rd AAAI Conference on Artificial Intelligence (AAAI-19)*.
- **Zhibing Zhao** Haoming Li, Junming Wang, Jeffrey Kephart, Nicholas Mattei, Hui Su, and Lirong Xia, "A Cost-Effective Framework for Preference Elicitation and Aggregation", in *Proceedings of the 34th Conference on Uncertainty in Artificial Intelligence (UAI-18)*.
- **Zhibing Zhao** and Lirong Xia, "Composite Marginal Likelihood Methods for Random Utility Models", in *Proceedings of the 35th International Conference on Machine Learning (ICML-18)*.
- **Zhibing Zhao**, Tristan Villamil, and Lirong Xia, "Learning Mixtures of Random Utility Models", in *Proceedings of the 32nd AAAI Conference on Artificial Intelligence (AAAI-18)*.
- **Zhibing Zhao**, Peter Piech, and Lirong Xia, "Learning Mixtures of Plackett-Luce Models", in *Proceedings of the 33rd International Conference on Machine Learning (ICML-16)*.
- Taofeek Orekan, **Zhibing Zhao**, Peng Zhang, Jian Zhang, Shengli Zhou, and Jun-Hong Cui, "Maximum Lifecycle Tracking for Tidal Energy Generation System", *Electric Power Components and Systems*, 2015.
- Gengfeng Li, Peng Zhang, Peter B. Luh, Wenyan Li, Zhaohong Bie, Camilo Serna, and **Zhibing Zhao**, "Risk Analysis for Distribution Systems in the Northeast U. S. Under Wind Storms", *IEEE Transactions on Power Systems*, 2014.
- **Zhibing Zhao**, Peng Zhang, Jun-Hong Cui, and Shengli Zhou, "Life-Oriented Control of Tidal Power Generation", *OCEANS*, 2013.

Professional Service

Reviewer for ICML-2020, NeurIPS-2020, PC member for AAAI-19, 20, reviewed multiple papers for Journal of Machine Learning Research, Int. J. of Renewable Energy Technology, etc.